

Project Documentation

Auto Ticket Classification using Flow Designer

Team ID:	SWTID-2026-5797
Date	3 october 2026

1. Introduction

- **Project Title:** Auto Ticket Classification using Flow Designer
- **Team Members:** Lavanya M A, Arthi L S, Harini T K, Karthiga M

2. Project Overview

Purpose: A school IT helpdesk receives multiple incident requests daily from students and teachers (Wi-Fi issues, projector failures, password problems, slow computers), and IT staff review each one manually to assign a category. This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the short description and description fields. Goals: automatically classify incidents at creation, reduce manual effort for IT agents, improve ticket routing efficiency, and implement a no-code, easily maintainable solution.

Features:

- Automatic Category and Subcategory assignment from keywords in the short description.
- Dependent choice logic between Category and Subcategory.
- Automated email notification to the caller upon ticket creation.
- Custom table with auto-numbering (INC prefix) storing tickets in a structured, standardized format.
- No-code, maintainable and scalable Flow Designer solution; transferable as an update set (XML).

3. Architecture

The project is a ServiceNow low-code/no-code solution and does not use the MERN stack; the template's three headings are answered with their ServiceNow equivalents.

Frontend

React is not used. The user interface is the ServiceNow Incident WorkFlow form, configured in Form Design (fields: Number, Caller, Category, Subcategory, Short description, Description, State, Assigned group, Assigned to). The student only enters Caller and Short description.

Backend

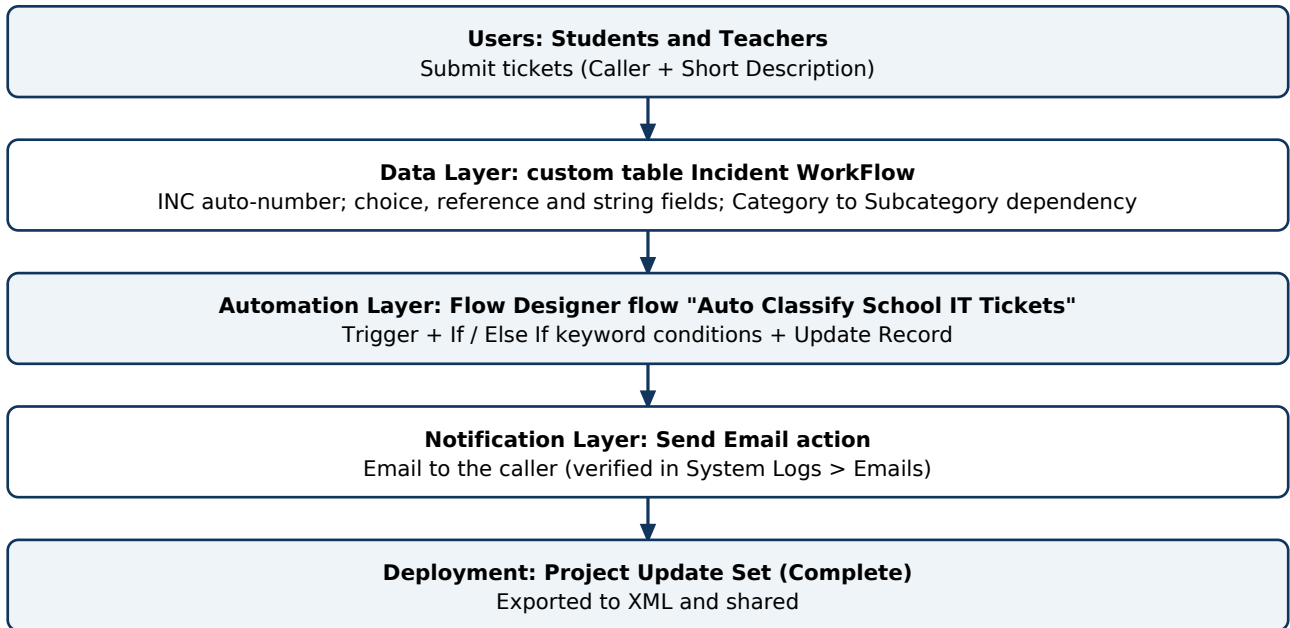
Node.js / Express.js are not used. The backend logic is the Flow Designer flow **Auto Classify School IT Tickets** (Global scope): trigger Record Created on Incident Workflow where Category is empty; If / Else If conditions on Short description "contains"; an Update Record action per branch; and a Send Email action to the caller.

Database

MongoDB is not used. Data is stored in the ServiceNow custom table **Incident WorkFlow** (u_incident_workflow), with auto-numbering (prefix INC, 5 digits). Schema:

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user

Field Label	Type	Reference / Choice
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice (dependent on Category)	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	—
Description	String	—
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user



4. Setup Instructions

Prerequisites

- A ServiceNow instance with access to System Update Sets, Tables, Form Design and Flow Designer (Workflow Studio).
- Email sending configured so the Send Email action can notify callers (emails were verified in System Logs → Emails).

Installation (configuration steps)

1. Create a Local Update Set named **Project Update Set** (State: In progress, Application: Global) and click Make This My Current Set.
2. Create the table **Incident WorkFlow** (uncheck Create module); on the Controls tab enable Auto-number (Prefix INC, Number 500, 5 digits).
3. In Form Design add the fields and choice values for Category, Subcategory and State.
4. Configure Dictionary on Subcategory → Advanced view → Dependent Field: Use dependent field, Dependent on field = Category; set each choice's Dependent value (Wi-Fi → network, Projector → hardware, Forgot password → access, Slow computer → performance).
5. In Flow Designer create the flow **Auto Classify School IT Tickets** (Global): trigger Record Created, table Incident Workflow, condition Category is empty.
6. Add If / Else If branches with Update Record actions (see section 6), add the Send Email action, then click **Activate**.
7. To share the work, set the update set State to Complete and use Export to XML.

5. Folder Structure

There is no client / server code base. The equivalent organization of the solution is:

Client (user interface)

- Incident WorkFlow form and Form Design layout (2-column default view).
- Category / Subcategory dependent choice lists.

Server (platform configuration)

- Table definition u_incident_workflow with auto-number and access controls (user role u_incident_workflow_user).
- Dictionary entries and choices (Category, Subcategory, State).
- Flow “Auto Classify School IT Tickets” – trigger, If / Else If logic, Update Record and Send Email actions.
- Project Update Set – packages all of the above (exported XML: sys_remote_update_set_a7b40de6fc590ceaad36c.xml, 182 KB).

6. Running the Application

There are no npm start commands. The application runs inside ServiceNow:

- **Start:** open the flow in Flow Designer and click **Activate** (top right corner).
- **Use:** open a new Incident WorkFlow record, enter Caller and Short description, then save and reload the form to see Category and Subcategory filled automatically.

Condition (Short description contains)	Category	Subcategory
Wi-Fi OR Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

7. API Documentation

The project exposes no REST endpoints. Its interface is the flow trigger and actions (no request methods or parameters beyond the following):

Element	Inputs / parameters	Result (example)
Trigger – Record Created	Table: Incident Workflow; condition: Category is empty	Flow starts for each new ticket.
Update Record (per branch)	Record: trigger record; Table: Incident WorkFlow; Fields: Category, Subcategory	INC00511: Category Network, Subcategory Wi-Fi.
Send Email	Target record; To: Caller → Email; Subject: “Your Request for the issue has been submitted.”; Body: ticket confirmation message	Email logged in System Logs → Emails (send-ready, SMTP).

8. Authentication

- Login and sessions are handled by the ServiceNow platform; the project document does not describe tokens or custom authentication.
- Authorization: the table is created with access controls (Create access controls, user role **u_incident_workflow_user**) so that only authorized users can access or modify ticket data; security validation is part of the testing phase.

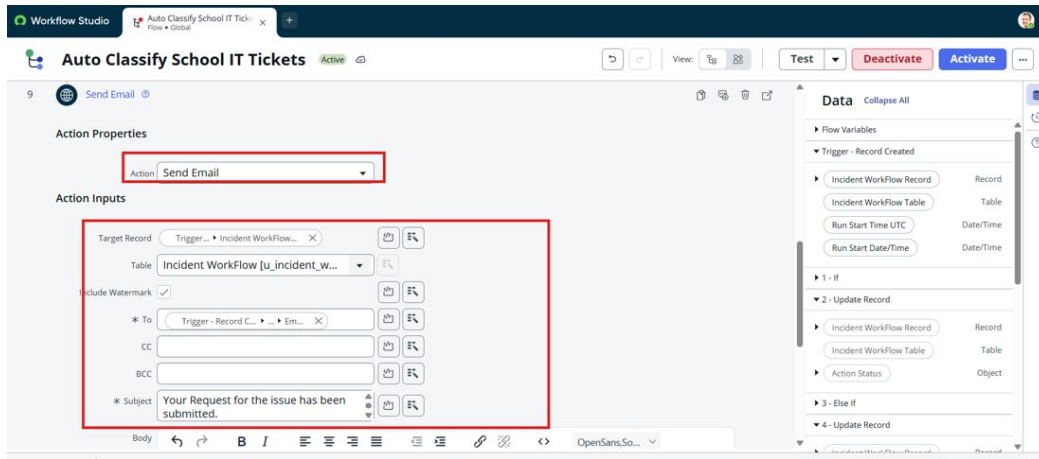
9. User Interface

Incident Workflow form: INC00511 with Category and Subcategory filled automatically.

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-02-06 21:45:02	access
Wi-Fi	wi-fi	en	0	false	2026-02-06 21:44:50	network
Slow Computer	slow computer	en	3	false	2026-02-06 21:45:15	performance
Projector	projector	en	1	false	2026-02-06 21:45:36	hardware

Subcategory choices with dependent values (Category-based filtering).

Flow Designer trigger: Record Created where Category is empty.



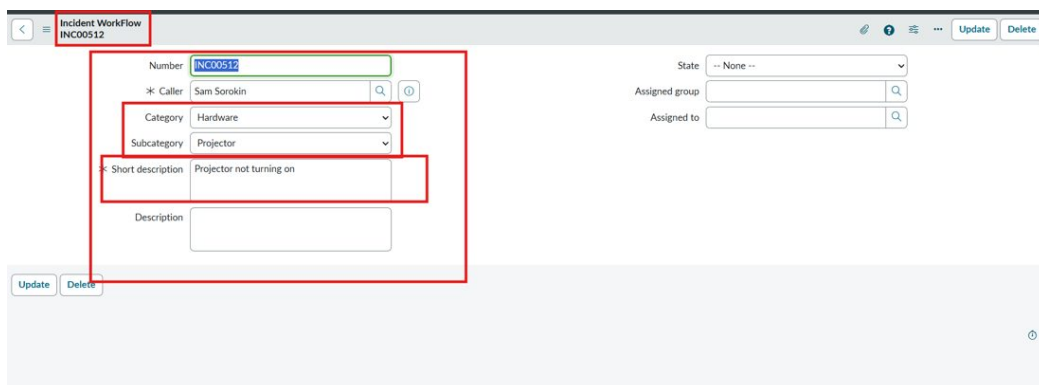
Send Email action configuration.

10. Testing

Strategy: manual functional validation of the end-to-end automation – data validation checks (mandatory fields, auto-number without duplication, accurate Category / Subcategory, reference fields resolving to user and group tables), two test scenarios, and an email check. Tools: ServiceNow forms, Flow Designer, and System Logs → Emails (Preview Email).

Test	Input	Result
Scenario 1	Short description “WiFi not working in library”	Network / Wi-Fi (INC00511); email sent to caller – Pass
Scenario 2	Short description “Projector not turning on”	Hardware / Projector (INC00512); email sent to caller – Pass
Email check	Search subject in System Logs → Emails; Preview Email	Confirmation email records and preview displayed – Pass
Not shown in project document	Password and Slow computer branches, empty mandatory fields, response time, concurrency	Not tested

11. Screenshots or Demo

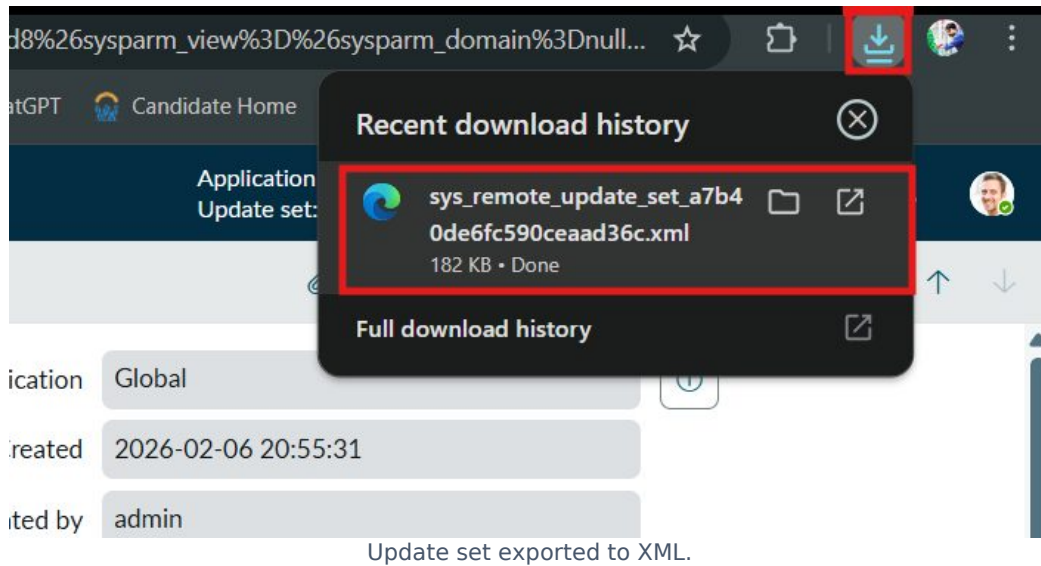


INC00512 auto-classified as Hardware / Projector.



Confirmation emails in System Logs → Emails.

Preview of the email sent to the caller.



Demo link: not provided.

12. Known Issues

- The overview lists the keywords “Password or Login” and “Slow or Hanging”, but the flow as built checks only “Forgot password” and “Slow computer”.
- Classification is keyword-based on the short description; other issue types are not auto-classified.
- Response time, concurrent load and the Password / Slow computer branches were not tested in the project document.

13. Future Enhancements

- Assignment automation.
- SLA tracking.
- Predictive intelligence.